

Directional Derivatives — Motivation

We already know two rates of change for $f(x, y)$:

- f_x = rate of change in the **x-direction** (holding y fixed)
- f_y = rate of change in the **y-direction** (holding x fixed)

But what if we want the rate of change in some other direction?

For example, northwest? At a 45° angle? Along a contour line?

The **directional derivative** $D_{\vec{u}}f$ measures the rate of change of f in the direction of any unit vector $\vec{u}$.

In fact, f_x and f_y are just special cases:

- $f_x(x, y) = D_{\vec{i}}f(x, y)$ — directional derivative in the $\vec{i}$ direction
- $f_y(x, y) = D_{\vec{j}}f(x, y)$ — directional derivative in the $\vec{j}$ direction

Definition of the Directional Derivative

Directional Derivative

Let $\vec{u} = \langle a, b \rangle$ be a **unit vector** ($|\vec{u}| = 1$). The **directional derivative** of f at (x, y) in the direction of $\vec{u}$ is:

$$D_{\vec{u}}f(x, y) = \lim_{t \rightarrow 0} \frac{f(x + ta, y + tb) - f(x, y)}{t}$$

provided this limit exists.

Reading the formula:

- Start at (x, y)
- Move a tiny step t in the direction $\vec{u} = \langle a, b \rangle$, arriving at $(x + ta, y + tb)$
- Measure the change in f , divide by t , take the limit

This is the same **rise-over-run** idea from Calc I, but now we walk in the direction $\vec{u}$ instead of along the x -axis.

Example 1: $D_{\vec{u}}f$ from the Definition

Example 1

Let $f(x, y) = x^2 + 4y^2$ and $\vec{u} = \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$.

Find $D_{\vec{u}}f(2, 1)$ using the **limit definition**.

Substitute into the limit definition. Expand, simplify, then take the limit.

The Gradient Vector

Before we find the shortcut, we need a new object.

Gradient Vector

If f has partial derivatives f_x and f_y , the **gradient** of f is the vector:

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = f_x \vec{i} + f_y \vec{j}$$

The symbol ∇ is called **nabla** or **del**.

The gradient packages **both partial derivatives** into a single vector. It lives in the **domain** (the xy -plane), not on the surface.

Example 2

Find ∇f for $f(x, y) = x^2 + 4y^2$. Then evaluate $\nabla f(2, 1)$.

Compute both partial derivatives, then evaluate at (2,1).

The Big Theorem: $D_{\vec{u}}f = \nabla f \cdot \vec{u}$

Theorem: Directional Derivative via the Gradient

If f is a differentiable function of x and y , then:

$$D_{\vec{u}}f(x, y) = \nabla f(x, y) \cdot \vec{u}$$

for any unit vector $\vec{u}$.

Translation: To find the rate of change in any direction, just take the **dot product** of the gradient with the direction vector.

This is the **central formula** of Section 14.6. It connects direction, rate of change, and gradient in one equation.

Why is this so much better than the definition?

- Limit definition: expand, simplify, take a limit (messy algebra)
- Gradient formula: compute two partial derivatives, take a dot product (fast!)

Example 3: $D_{\vec{u}}f$ via the Gradient

Example 3

Same problem as Example 1: $f(x, y) = x^2 + 4y^2$, $\vec{u} = \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$.

Find $D_{\vec{u}}f(2, 1)$ using the gradient.

Use the formula: gradient dot direction.

Maximizing the Directional Derivative

Question: You're standing at a point (a, b) in the domain of f . Which direction should you move to increase f as fast as possible?

Using the dot product identity:

$$D_{\vec{u}}f = \nabla f \cdot \vec{u} = |\nabla f| |\vec{u}| \cos \theta = |\nabla f| \cos \theta$$

where θ is the angle between ∇f and $\vec{u}$.

The **only thing we can control** is the angle θ . The gradient ∇f is fixed at each point.

So maximizing $D_{\vec{u}}f$ comes down to maximizing $\cos \theta$:

- $\cos \theta = 1$ when $\theta = 0$ — $\vec{u}$ points **with** ∇f → **maximum** rate = $|\nabla f|$
- $\cos \theta = -1$ when $\theta = \pi$ — $\vec{u}$ points **against** ∇f → **minimum** rate = $-|\nabla f|$
- $\cos \theta = 0$ when $\theta = \pi/2$ — $\vec{u}$ is **perpendicular** to ∇f → rate = 0

Summary: Three Key Cases

Condition	Angle θ	$D_{\vec{u}}f$	Meaning
$\vec{u}$ parallel to ∇f	0	$ \nabla f $ (max)	Fastest increase
$\vec{u}$ opposite to ∇f	π	$- \nabla f $ (min)	Fastest decrease
$\vec{u} \perp \nabla f$	$\pi/2$	0	No change (along level curve)

The gradient tells you everything:

- Its **direction** = direction of steepest ascent
- Its **magnitude** = the maximum rate of change
- **Perpendicular** to it = along the level curve (no change in f)

Important clarification: The gradient vector ∇f lives in the **domain** (the xy -plane). It is NOT a vector "pointing uphill on the surface." It points toward larger f -values in the input space.

Example 4: Fastest Increase and Decrease

Example 4

For $f(x, y) = x^2 + 4y^2$ at the point $(2, 1)$:

- (a) What direction gives the **fastest increase** in f ? How fast?
- (b) What direction gives the **fastest decrease**?
- (c) In what directions is $D_{\vec{u}}f = 0$?

Use the gradient and the max/min/zero cases from the table.

Example 5: Predicting Change with the Gradient

Example 5

Continued from Example 4: If you travel a distance of 0.1 from $(2, 1)$ in the direction of ∇f , predict the change dz in $z = f(x, y)$.

The maximum rate of change times the distance traveled gives dz .

Your Turn: Directional Derivative Practice

Example 6

Let $f(x, y) = xe^{-y}$ and let $\vec{u}$ be the unit vector in the direction of $\langle 2, -1 \rangle$.

- (a) Find $\nabla f(2, 0)$.
- (b) Find $D_{\vec{u}}f(2, 0)$.
- (c) What is the maximum rate of change of f at $(2, 0)$?

First normalize the direction vector. Then compute the gradient, evaluate, and take the dot product.

Gradient Is Perpendicular to Level Curves

Theorem: Gradient Perpendicular to Level Curves

The gradient vector $\nabla f(a, b)$ is **perpendicular** to the level curve $f(x, y) = k$ at the point (a, b) .

Why does this make sense?

Along a level curve, f is constant, so the rate of change is **zero**. We saw that $D_{\vec{u}}f = 0$ when $\vec{u} \perp \nabla f$.

The tangent direction to a level curve is a direction of zero change — so it must be perpendicular to ∇f .



Contour map of $f(x,y) = x^2 + 4y^2$ showing elliptical level curves with the gradient vector at $(2,1)$ pointing perpendicular to the level curve

[Desmos 3D](https://www.desmos.com/3d) [https://www.desmos.com/3d]

The gradient always points **perpendicular to level curves**, aiming toward higher values of f . This is why contour maps and gradients are deeply connected.

Example 7: Tangent Line to a Level Curve

Example 7

Consider $f(x, y) = x^2 + 4y^2$ with level curves that are ellipses.

At the point $(2, 1)$, find the equation of the **tangent line** to the level curve $f(x, y) = 8$.

The gradient is normal to the level curve. Use ∇f as the normal to write the tangent line equation.

Extending to Three Variables

Everything generalizes naturally to functions of three variables $f(x, y, z)$:

Gradient in Three Variables

$$\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle$$

The directional derivative formula is exactly the same:

$$D_{\vec{u}}f(x, y, z) = \nabla f(x, y, z) \cdot \vec{u}$$

where $\vec{u} = \langle a, b, c \rangle$ is a unit vector in $\mathbb{R}^3$.

Level surfaces replace level curves:

For $f(x, y, z) = k$, the gradient ∇f is perpendicular to the **level surface** at each point.

Since ∇f is perpendicular to the level surface, it serves as the **normal vector** for the tangent plane!

Tangent plane to $f(x, y, z) = k$ at (a, b, c) :

$$f_x(a, b, c)(x - a) + f_y(a, b, c)(y - b) + f_z(a, b, c)(z - c) = 0$$

Example 8: Tangent Plane to a Level Surface

Example 8

Let $f(x, y, z) = zx^2 - yz^2$.

(a) Compute $\nabla f(1, 2, 3)$.

(b) Find the equation of the tangent plane to the level surface $f(x, y, z) = -15$ at the point $(1, 2, 3)$.

Find all three partials, evaluate at $(1, 2, 3)$, then use the tangent plane formula.

Homework

Section 14.6: 1, 5, 9, 11, 13, 19, 21, 29, 31, 35, 39, 44, 49, 51, 55, 61, 65, 73